



LIS INTERFACE INSTRUCTION MANUAL

Goldsite Diagnostics Inc.

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Foreword

Before using this system, please read and understand the contents of this manual in order to use the system correctly. Save this manual after reading and place this manual in a convenient and proper location.

Readers of Manual

The readers of this manual are those developing LIS and those who need to know the HL7 interface of Goldsite. This manual is used to guide the LIS system developer to develop the LIS interface so that its LIS system can communicate with the instrument for data transmission. Developers need to have certain LIS, HL7 standards related knowledge and network programming capabilities. The communication protocol of the network is TCP/IP.

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HL7 Interface

Introduction

The HL7 interface is a new feature developed by Goldsite to accommodate today's fast-developing Laboratory Information System (LIS), providing a gateway to Goldsite's instruments and other IP-based networks. Using a TCP/IP connection, the LIS server can receive test data from the instrument. The instrument sends the information of patient, sample and test results, etc. to the LIS server in the real-time and batch transmission manner.

Supported Messages

HL7 is an electronic data exchange standard that provides care for inpatients. Originally defined by the United States, it has now been adopted by many countries. The interface described in this manual is defined based on HL7 v2.3.1. For details, please refer to HL7 Interface Standards Version 2.3.1.

Because the Goldsite's HL7 interface only uses part of the data, only a subset of the message types, segment types and other data specified in HL7 are used in this interface.

Bottom Layer Protocol

TCP/IP is a byte stream protocol that does not provide message boundaries. As an upper layer protocol, HL7 is message-based, but it does not provide a message termination mechanism. In order to determine the message boundary, we use the smallest bottom layer protocol (as described in HL7 Interface Standards Version 2.3.1).

The messages in the communication layer are transmitted in the following format:

<SB> ddddd <EB><CR>

Interpretation:

<SB> Interpretable: Start Block character (1 byte), ASCII <VT>, ie, <0x0B>. Do not confuse with the ASCII characters SOH or STX.

dddd Interpretable: Data (variable number of bytes), dddd is an HL7 message. Only ISO 8859-1 characters (hexadecimal values 20 - FF) and <CR> are included. The other control and unprintable characters are not included.

<EB> Interpretable: End Block character (1 byte), ASCII <FS>, ie, <0x1C>. Do not confuse with the ASCII characters ETX or EOT.

<CR> Interpretable: Carriage Return (1 byte), ASCII carriage return, ie, <0x0D>.

Duplex Communication

The following aspects are included:

1. Transmission of test results

The transmission of test results mainly enables the instrument to transmit the test data to an external system (such as LIS). The data transmission methods include the real-time transmission during testing and the batch transmission of previous results. It is stipulated in the system settings that the real-time transmission is enabled during testing; that the batch transmission cannot be enabled during testing but in other cases, batch transmission can be performed.

2. Acquisition of sample application information

This refers to the acquisition of the sample application information from the LIS for the local instrument to perform the test. The acquisition methods can be real-time transmission (used only by the instrument equipped with the sample scanning system; acquisition by scanning the barcodes of samples) and batch acquisition (sample information in the time period from the beginning of the day to the request time).

3. If the samples have been accepted, and the results are not available yet, they will be sent by the DSP segment.

4. If the samples have been accepted, and all the respective results tested by the instrument have already been available, they will be sent by the ERR segment.

5. If the item barcodes of the samples cannot be found in the instrument, they will be sent by the ERR segment, the same as described above in point 4.

Introduction

This section describes several HL7 (version 2.3.1) messages used by the Goldsite HL7 interface.

Note:

HL7 supports many message types, only three messages are used here.

1. Sample test results
2. Quality control results
3. Calibration results

Message Syntax

Note: For a complete and detailed description of the HL7 message syntax, please refer to the HL7 standard provided by the HL7 Standards Committee. Each HL7 message consists of segments that end with <CR>. Each segment consists of a three-character segment name and fields with variable numbers. The domain consists of components and subcomponents, and the separators for each component are defined in the MSH section of each message.

Example:

MSH|^~\&|GX|AA|||20181127194802||ORU^R01|1|P|2.3.1|||0||ASC|||

Interpretation:

The five-character definition after MSH is used to distinguish the separators for each domain, component, and subcomponent. Although these characters can be any non-text characters, the following characters are recommended by the HL7 standard:

Character	Description
	Domain separator
^	Component separator
&	Subcomponent separator
~	Repeat separator
\	Escape character

The first field of MSH includes the various separators. Some of the fields that follow are empty because they are optional and not used by Goldsite HL7 interface. The detailed definition and selection of the fields are described below.

Domain 9: Contains message types and events (ORU, R01)

Domain 10: Contains a message ID that uniquely identifies the message

Domain 11: Contains the processing ID (P for product)

Domain 12: Defines the HL7 version (2.3.1)

For any type of message, the order of the segments after the MSH segment is specified. These sequences are described in detail in the following sections. The syntax structures below are used to indicate segment that is optional or repeatable.

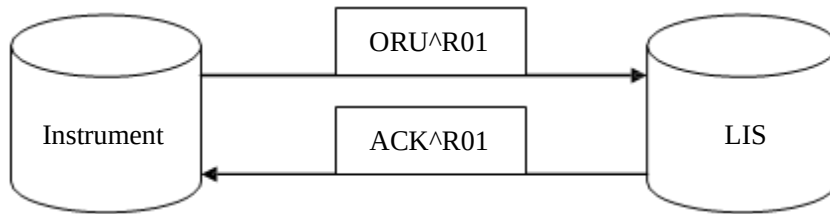
[] indicates that the segment inside is optional.

{ } means that each segment can be repeated 0 times or 1 or more times.

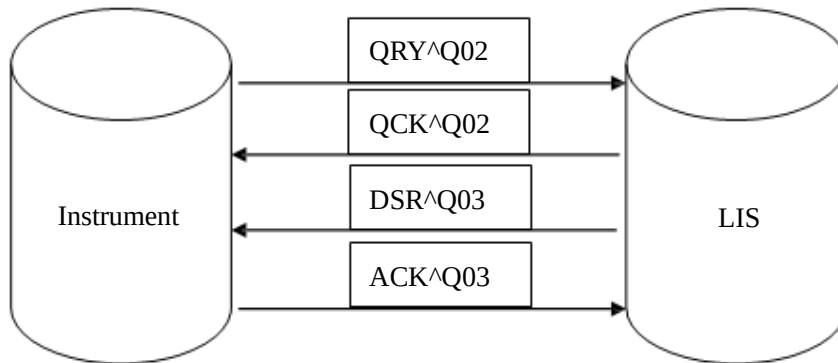
Supported HL7 messages

The HL7 messages used by this interface are ORU, ACK, QRY, QCK, DSR.

The upload of the test results is as follows:



The acquisition of sample application information from the LIS server is as follows:



Interpretation:

ORU/ACK: Unsolicited observation report/response

The primary role of the ORU^R01 message is to transmit laboratory results in HL7. This is used by Goldsite to transmit patient sample test results, calibration results, and quality control results to the LIS system.

For the test results of patient samples, the following information are included:

- Patient information (patient name, sample number, etc.)
- Medical information (sample type, physician, examiner, clinical diagnosis, etc.)
- Test result (a group of messages, each of which corresponds to a sample, and there may be results for multiple items for one sample). The LIS system can choose to use these information as needed.

The structure is as follows:

ORU Observational Results (Unsolicited)	
MSH	Message header
PID	Patient identification information
OBR	Observation report
{OBX}	Test result

For the results of calibration, the following information are included:

- Item information (item number, item name)
- Calibration rules, calibration time
- Calibrator related information (number, ID, name, lot number, expiration date, concentration, etc.)
- Calibration results (degree of reaction, number of calibration parameters, calibration parameter value). A message will send all calibration results for an item. The LIS system can choose to use these information as needed.

For the results of quality control testing, the following information are included:

- Item information (item number, item name)
- Quality control material related information (number, ID, name, lot number, expiration date, mean value)
- QC time, test results (concentration value)

A message will send all quality control test results for an item. The LIS system can choose to use these information as needed.

The transmission structure of the calibration and quality control results is as follows:

ORU Observational Results (Unsolicited)	
MSH	Message header
OBR	Observation report

The ACK^R01 message is used to respond to the ORU message. The structure is as follows:

ACK Acknowledgement	
MSH	Message header
MSA	Message confirmation

QRY/QCK: Query observations/responses

The QRY^Q02 message queries the current data and is used to query the LIS system for the required sample application information, which triggers event Q02. The structure is as follows:

QRY Query	
MSH	Message header
QRD	Query definition
QRF	Query filter

The QCK^Q02 message is used to respond to QRY messages. The structure is as follows:

QCK Query Acknowledgement	
MSH	Message header
MSA	Message confirmation
ERR	Error message
QAK	Query confirmation

DSR/ACK: observation result display/response

The main function of the DSR^Q03 message is to send and display of the query result, that is, transmission of the required sample application information from LIS to the instrument. According to the HL7 standard, the following structure is used:

DSR Display Response

MSH	Message header
MSA	Message confirmation
ERR	Error message
QAK	Query confirmation
QRD	Query definition
QRF	Query filter
{DSP}	Display data
DSC	Continuous pointer

The ACK^Q03 message is used to respond to the DSR message. The structure is as follows:

ACK Acknowledgement	
MSH	Message header
MSA	Message confirmation
ERR	Error message

Message Segment

Note:

In the Goldsite HL7 interface, none of the fields in the message segment are omitted for future expansion needs, and if the field has no value, it is blanked.

The Goldsite HL7 interface includes (but not limited to) the MSH segment, PID segment, OBR segment, OBX segment, QRD segment, QRF segment, MSA segment, DSP segment, ERR segment. Currently for development purpose of Goldsite HL7, the MSH segment, PID segment, OBR segment, OBX segment, QRD segment, QRF segment, MSA segment, DSP segment, and ERR segment should be understood.

For the first message segment of the HL7 message, all messages start with the MSH segment, usually at the top of the message. This message segment is used to define the message's intent, source, purpose, and some details of the message syntax.

MSH Segment

No.	Field	Length	Description
1	Field Separator	1	Contains the separator between the segment ID and the first real domain, defining the domain separator () for the rest of the message
2	Encoding Characters	4	Contains component delimiters, repeat delimiters, escape delimiters, and subcomponent separators (^~\&)
3	Sending Application	180	Sending application
4	Sending Facility	180	Sending device, by the model
5	Receiving Application	180	Blank
6	Receiving Facility	180	Blank
7	Date/Time Of Message	26	Time of current message, time information for system calling
8	Security	40	Blank
9	Message Type	7	Message type, e.g., ORU^R01

10	Message Control ID	20	Message control ID, uniquely identifies a message, increments from 1 as the number of messages increases
11	Processing ID	3	Processing ID, always uses P (for product)
12	Version ID	60	HL7 version, 2.3.1
13	Sequence Number	15	Blank
14	Continuation Pointer	180	Blank
15	Accept Acknowledgment Type	2	Valid only for DSR^Q03 messages. The value is P indicating the sample information sent by the LIS system. Other values indicate sample information in response to a LIS query and can be made blank.
16	Application Acknowledgment Type	2	Type of the results: 0 – sample test result; 1 – calibration result; 2 – quality control result
17	Country Code	2	Blank
18	Character Set	10	Uses ASCII characters
19	Principal Language Of Message	60	Blank
20	Alternate Character Set Handling Scheme	20	Blank

Note: This section will appear in all messages. In the HL7 message sent by the LIS system to the instrument, the values of the fields 3 and 4 of the segment are determined by the LIS developer, and the fields 5 and 6 are "GOLSITE" and the device model, respectively. The fields 10 and 16 use integer data type; and all other fields use string data type. When the LIS system actively sends the sample application to the instrument, the field 15 needs to be value "P".

PID Segment

No.	Field	Length	Description
1	Set ID – PID	10	Identifies different patients
2	Fixed	1	Fixed to 1
3	Patient ID	20	Blank
4	Patient Identifier List	20	Blank
5	Alternate Patient ID – PID	20	Blank
6	Patient Name	48	Patient's name
7	Mother's Maiden Name	48	Blank
8	Date/Time of Birth	26	Date of birth of the patient (format: year, month, day, hour, minute, second).

			19830512000000), can be empty.
9	Sex	1	M for male; F for female; O for others; U for unknown
10	Patient Alias	48	Blood type ('A' or 'B' or 'AB' or 'O' for the respective blood types, other values might be used as defined by the customers)
11	Race	80	Blank
12	Patient Address	106	Blank
13	County Code	4	Blank
14	Phone Number - Home	40	Blank
15	Phone Number -Business	40	Blank
16	Primary Language	60	Blank
17	Marital Status	80	Blank
18	Religion	80	Blank
19	Patient Account Number	20	Blank
20	SSN Number-Patient	16	Blank
21	Driver's License Number – Patient	25	Blank
22	Mother's Identifier	20	Blank
23	Ethnic Group	80	Blank
24	Birth Place	60	Blank
25	Multiple Birth Indicator	1	Blank
26	Birth Order	2	Blank
27	Citizenship	80	Blank
28	Veterans Military Status	60	Blank
29	Nationality	80	Blank
30	Patient Death Date and Time	26	Blank
31	Patient Death Indicator	1	Blank

Note: This message segment is only used in the ORU^R01 message. The first and second fields use integer data type; the second and third fields use boolean data type; others use string data type.

OBR Segment

The OBR segment is used to transmit medical information related to the test report. For transmission of the test result information of patient samples (MSH-16 is 0).

No.	Field	Length	Description
1	Set ID–OBR	10	Identifies different OBR fields
2	Sample Code	22	Sample number
3	Filler Order Number	22	Test date + sample number
4	Universal Service ID	200	Blank
5	Sending Facility	180	Sending device, by the model
6	Priority	2	Whether it is STAT, Y for yes, N for no
7	Requested Date/Time	26	Requested date/time. Used as sample collection time
8	Observation Date/Time	26	Observation date/time. Used as inspection time
9	Observation End Date/Time	26	End of the observation date/time
10	Collection Volume	20	Blank
11	Collector Identifier	60	Blank
12	Specimen Action Code	1	Blank
13	Danger Code	60	Blank
14	Relevant Clinical Info	300	Blank
15	Specimen Received Date/Time	26	Blank
16	Specimen Source	300	Used as sample type, can be serum or plasma or urine, other types might be used as defined by the software
17	Ordering Provider	120	Blank
18	Order Callback Phone Number	40	Blank
20	Placer Field 1	60	Blank
21	Placer Field 2	60	Blank
22	Filler Field 1	60	Blank
23	Filler Field 2	60	Blank
24	Result Rpt/Status Change –Date/Time	26	Blank
25	Charge to Practice	40	Blank
26	Diagnostic Serv Sect ID	10	Blank
27	Result Status	1	Blank
28	Parent Result	200	Blank
29	Quantity/Timing	200	Blank
30	Result Copies To	150	Blank
31	Parent	150	Blank
32	Transportation Mode	20	Blank
33	Reason for Study	300	Blank
34	Principal Result Interpreter	200	Blank
35	Assistant Result	200	Blank

	Interpreter		
36	Technician	200	Blank
37	Transcriptionist	200	Blank
38	Scheduled Date/Time	26	Blank
39	Number of Sample Containers	4	Blank
40	Transport Logistics of Collected Sample	60	Blank
41	Collectors Comment	200	Blank
42	Transport Arrangement Responsibility	60	Blank
43	Transport Arranged	30	Blank
44	Escort Required	1	Blank
45	Planned Patient Transport Comment	200	Blank
46	Ordering Facility Name	60	Blank
47	Ordering Facility Address	106	Blank
48	Ordering Facility Phone Number	48	Blank
49	Ordering Provider Address	106	Blank

OBX Segment

OBX is primarily used to transmit the observed information in report messages. If the patient's sample test information is transmitted (MSH-16 is 0) - a patient may have multiple OBXs, this interface does not provide repetitive detection of the test data, and the data shall be processed by the user's own system.

No.	Field	Length	Description
1	Set ID –OBX	10	Identifies different OBX fields
2	Fixed	1	Fixed to 1
3	Observation Identifier	590	Used as item ID
4	Value Type	3	Value types, NM (numeric) for quantitative assay, ST (string) for qualitative assay
5	Observation Sub-ID	20	Used as item name
6	Observation Value	65536	Used as result value for quantitative assay (concentration)
7	Units	90	Used as the unit of test result
8	References Range	90	The expected normal range
9	Abnormal Flags	5	Indicates whether the result value falls in the normal range: L for low, H for high, N for normal
10	Probability	5	Blank

11	Nature of Abnormal Test	2	Blank
12	Observe Result Status	1	Uses F-final results
13	Date Last Observe Normal Values	26	Blank
14	User Defined Access Checks	20	Used as the original result
15	Date/Time of the Observation	28	Used as test time
16	Producer's ID	60	Blank
17	Responsible Observer	80	Blank
18	Observation Method	60	Debug field

Note: This message segment only appears in the ORU^R01 message. Fields 1, 3, and 9 use integer data type; fields 5 and 13 use floating-point data type; others use string data type.

QRD Segment

No.	Field	Length	Description
1	Query Date/Time	26	Time of the query, the system time is used
2	Query Format Code	1	Uses R for record-oriented format
3	Query Priority	1	Uses D for deferred
4	Query ID	10	Identifies different queries, increments from 1 as the number of queries increases
5	Deferred Response Type	1	Blank
6	Deferred Response Date/Time	26	Blank
7	Quantity Limited Request	10	Limitation of quantity, uses RD for records
8	Who Subject Filter	60	Filter by patient, uses patient's barcode
9	What Subject Filter	60	Filter by contents, sets to OTH when querying
10	What Department Data Code	60	Blank
11	What Data Code Value Qual	20	Blank
12	Query Results Level	1	Query for result level, uses T for Full results

Note: This message segment may appear in the QRY^Q02 message and the DSR^Q03

message. The field 8 is the sample barcode number when it is acquired in real time and is empty when it is acquired in batch. The field 4 uses the integer data type; and the other fields use the string data type.

QRF Segment

No.	Field	Length	Description
1	Where Subject Filter	20	Query filter by address, uses the instrument model
2	When Data Start Date/Time	26	Blank
3	When Data End Date/Time	26	Blank
4	What User Qualifier	60	Blank
5	Other QRY Subject Filter	60	Blank
6	Which Date/Time Qualifier	12	Target type, uses RCT (Specimen receipt Date/time, receipt of specimen in filling ancillary (Lab))
7	Which Date/Time Status Qualifier	12	Target status, uses COR (Corrected only (no Final with corrections))
8	Date/Time Selection Qualifier	12	Date/time selection qualifier, uses ALL (All values within the range)
9	When Quantity/Timing Qualifier	60	Blank

MSA Segment

No.	Field	Length	Description
1	Acknowledgment Code		Confirmation code, AA means acceptance; AE indicates an error; AR indicates a rejection
2	Message Control ID		Message Control ID, the same with sender MSH-10
3	Text Message		A textual description of the event when an error or a rejection is encountered. Corresponds to field 6. Can be used to write error logs
4	Expected Sequence Number		Blank
5	Delayed Acknowledgment Type		Blank
6	Error Condition		Error condition (status code)

For field 6, the status codes of the error conditions are shown in the following table.

Status code (MSA-6)	Text (MSA-3)	Description
Successful		AA
0	Message accepted	Successful
Error status code		AE
100	Segment sequence error	The order of the middle of the message is incorrect, or the required segment is missing.
101	Required field missing	Required field missing in one segment
102	Data type error	Data type error, e.g., the numbers become characters
103	Table value not found	Table value not found, which would not be used temporarily
Rejection status code		AR
200	Unsupported message type	Message type is not supported
201	Unsupported event code	Event code is not supported
202	Unsupported processing ID	Processing ID is not supported
203	Unsupported version ID	Version ID is not supported
204	Unknown key identifier	E.g., transmission of the information of a patient that does not exist
205	Duplicate key identifier	Duplicate key identifier
206	Application record locked	The event cannot be executed at the application storage level, e.g., the database is locked
207	Application internal error	Unknown internal error in application

Note: The message segment can appear in the ACK^R01, QCK^Q02, ACK^Q03 messages.

The fields 4 and 6 use the integer data type; other fields use the string data type. This segment is noted with DSP at the same time.

If the calibration result is transmitted (MSH-16 is 1), the fields are defined as follows:

No.	Field	Length	Description
1	Set ID - OBR	10	Identifies different OBR fields
2	Placer Order Number	22	Requester's medical number, used as item number
3	Filler Order Number	22	Executor's medical number, used as item

			name
4	Universal Service ID	200	Universal service identifier, uses vendor name^model
5	Priority	2	Blank
6	Requested Date/time	26	Blank
7	Observation Date/Time	26	Observation date/time, used as calibration time
8	Observation End Date/Time	26	End of the observation date/time
9	Collection Volume	20	Used as calibration rule: 0 - Single-point linear; 1- Two-point linear; 2 - Multi-point linear; 3 - Logistic-Log4P; 4 - Logistic-Log5P; 5 - Exponential 5P; 6 - Polynomial 5P; 7 - Parabola; 8 - Spline
10	Collector Identifier	60	Blank
11	Specimen Action Code	1	Number of calibrators
12	Danger Code	60	Calibrator ID
13	Relevant Clinical Info	300	Name of calibrator
14	Specimen Received Date/Time	26	Lot of calibrator
15	Specimen Source	300	Expiration date of calibrator
16	Ordering Provider	120	Standard concentration of calibrator
17	Order Callback Phone Number	40	Concentration level of calibrator: H for high, M for middle, L for low
18	Placer Field 1	60	Degree of reaction
19	Placer Field 2	60	Number of calibration parameters
20	Filler Field 1	60	The values of the calibration parameters
21	Filler Field 2	60	Blank
22	Result Rpt/Status Change - Date/Time	26	Result report/status change – date/time
23	Charge to Practice	40	Blank
24	Diagnostic Serv Sect ID	10	Blank
25	Result Status	1	Blank
26	Parent Result	200	Blank
27	Quantity/Timing	200	Blank
28	Result Copies To	150	Blank
29	Parent	150	Blank
30	Transportation Mode	20	Blank
31	Reason for Study	300	Blank

32	Principal Result Interpreter	200	Blank
33	Assistant Result Interpreter	200	Blank
34	Technician	200	Blank
35	Transcriptionist	200	Blank
36	Scheduled Date/Time	26	Blank
37	Number of Sample Containers	4	Blank
38	Transport Logistics of Collected Sample	60	Blank
39	Collectors Comment	200	Blank
40	Transport Arrangement Responsibility	60	Blank
41	Transport Arranged	30	Blank
42	Escort Required	1	Blank
43	Planned Patient Transport Comment	200	Blank
44	Ordering Facility Name	60	Blank
45	Ordering Facility Address	106	Blank
46	Ordering Facility Phone Number	48	Blank
47	Ordering Provider Address	106	Blank

Note: This message segment is only used in the ORU^R01 message. The fields 1, 9, 11, 19, 37 use the integer data type; the field 10 uses the floating-point data type; the other fields use the string data type. The contents of the fields 12 - 18 are based on the number of calibrators, and the format is V1^V2^...Vi. The field 20 is based on the number of calibration parameters, and the different values are also separated by the ^ separator.

Different calibration rules have different calibration parameters.

For single-point linear, two-point linear, and multi-point linear, there are two calibration parameters K, R0;

For Logistic-Log4P, there are four calibration parameters K, R0, a, b;

For Logistic-Log5P and Exponential 5P, there are 5 calibration parameters K, R0, a, b, c;

For Polynomial 5P, there are 5 calibration parameters R0, a, b, c, d;

For Parabola, there are 3 calibration parameters R0, a, b;

For Spline, there are 4*(n-1) calibration parameters R0i, ai, bi, ci, where n is the number of calibrators.

If the result of the quality control testing is transmitted (MSH-16 is 2), the fields are defined as follows:

No.	Field	Length	Description
1	Set ID - OBR	10	Identifies different OBR fields
2	Placer Order Number	22	Requester's medical number, used as item number
3	Filler Order Number	22	Executor's medical number, used as

			item name
4	Universal Service ID	200	Universal service identifier, uses vendor name^model
5	Priority	2	Blank
6	Requested Date/Time	26	Blank
7	Observation Date/Time	26	Observation date/time, used as quality control name
8	Observation End Date/Time	26	End of observation date/time
9	Collection Volume	20	Blank
10	Collector Identifier	60	Blank
11	Specimen Action Code	1	Number of quality control
12	Danger Code	60	Quality control ID
13	Relevant Clinical Info	300	Name of quality control
14	Specimen Received Date/Time	26	Lot of quality control
15	Specimen Source	300	Expiration date of quality control
16	Ordering Provider	120	Blank, reserved
17	Order Callback Phone Number	40	Concentration level of quality control
18	Placer Field 1	60	Mean concentration of quality control
19	Placer Field 2	60	Standard deviation of quality control
20	Filler Field 1	60	Test result (concentration)
21	Filler Field 2	60	Blank
22	Result Rpt/Status Change -Date/Time	26	Result report/status change – date/time
23	Charge to Practice	40	Blank
24	Diagnostic Serv Sect ID	10	Blank
25	Result Status	1	Blank
26	Parent Result	200	Blank
27	Quantity/Timing	200	Blank
28	Result Copies To	150	Blank
29	Parent	150	Blank
30	Transportation Mode	20	Blank
31	Reason for Study	300	Blank
32	Principal Result Interpreter	200	Blank
33	Assistant Result Interpreter	200	Blank
34	Technician	200	Blank
35	Transcriptionist	200	Blank
36	Scheduled Date/Time	26	Blank
37	Number of Sample Containers	4	Blank
38	Transport Logistics of Collected Sample	60	Blank
39	Collector's Comment	200	Blank
40	Transport Arrangement Responsibility	60	Blank

41	Transport Arranged	30	Blank
42	Escort Required	1	Blank
43	Planned Patient Transport Comment	200	Blank
44	Ordering Facility Name	60	Blank
45	Ordering Facility Address	106	Blank
46	Ordering Facility Phone Number	48	Blank
47	Ordering Provider Address	106	Blank

Note: This message segment is only used in the ORU^R01 message. Fields 1, 11, and 37 use the integer data type; other fields use the string data type. The format of fields 12, 13, 14, 15, 17, 18, 19, and 20 are V1^V2^...Vi according to the number of quality controls. It should be noted that for daytime quality control, each quality control testing is transmitted in one message; while for real-time and between-day quality control, several quality control testing for each item is combined and transmitted in one message, but for different lots, the results are sent separately.

DSP Segment

No.	Field	Length	Description
1	Set ID - DSP	8	Fixed: 10000001
2	Display Level	1	Fixed: 1
3	Project Name ID	60	Item ID, multiple items can be separated by ","

Note: If the samples have been accepted, and the results are not available yet, they will be sent in the following format (barcode number 10000001, the items' ID are 51, 1, ,4, which correspond to bCRP, IgA, IgG, respectively).

ERR Segment

The ERR segment is used to add an error description to the acknowledgment message.

No.	Field	Length	Description
1	Error Code	1	Fixed, sends 0

Note: If the samples have been accepted, and all the test items belonging to the instrument already have the results or the loading barcode number cannot be found in the instrument, the format above will be transmitted.

Communication Process and Message Example

Format

<SB> ddddd <EB><CR>

Where <SB> indicates the beginning of the message, and the corresponding ASCII character is <VT>, which is 0x0B;

<EB> indicates the end of the message, and the corresponding ASCII character is <FS>, which is 0x1C;

<CR> indicates the confirmation of the end of message, and the separator between different messages, which is 0x0D;

dddddd is the actual content to be transmitted, including several segments with each segment ending with <CR>, which is 0x0D.

Example

The instrument sends the sample test results to the LIS server, which is transmitted in samples, that is, the tests contained for one sample is transmitted together as a message. After the LIS server receives and judges the message, it responds accordingly.

The ORU message contains segments of MSH, PID, OBR, and OBX (if a sample contains multiple items, there are multiple OBX segments). MSH is the message header and is included in any message. The format of the content is as follows (the text in parentheses in each field is the interpreted language, not the content of the field itself):

MSH|^~\&|GOLSITE (vendor name)|AA (device model)|||20181123101830 (system time, format yyyymmddhhMMss)||ORU^R01 (message type)|1 (message control ID, identifies message, increments from 1 as the number of messages increases)|P (fixed value, indicates the product)|2.3.1 (the version number of the HL7 protocol used)|||0 (0 for sample test result, 1 for calibration test result, 2 for quality control test result, blank for others)||ASCII (character set)|||

Note: In all the following message examples, in the field indicating time, if it has 14 digits, the time format is year (4 digits) month (2 digits) day (2 digits) hour (2 digits) minute (2 digits) second (2 digits); if it has 8 digits, the time format is year (4 digits) month (2 digits) day (2 digits).

The following page shows a sample sent by Lis for reference.

